Research Square

Preprints are preliminary reports that have not undergone peer review.
They should not be considered conclusive, used to inform clinical practice,
or referenced by the media as validated information.

Cultural Influences on Personality Types: A Cluster Analysis of HEXACO Traits in Japan

Eisuke Tokiwa

eisuke.tokiwa@sunblaze.jp

SUNBLAZE Co., Ltd. https://orcid.org/0009-0009-7124-6669

Article

Keywords: HEXACO model, personality types, cluster analysis, Japanese culture, person-centered

Posted Date: October 31st, 2024

DOI: https://doi.org/10.21203/rs.3.rs-5323735/v1

License: This work is licensed under a Creative Commons Attribution 4.0 International License.
Read Full License

Additional Declarations: There is **NO** Competing Interest.

Abstract

This study aims to identify personality types among Japanese individuals using the HEXACO model, employing cluster analysis based on over 13,000 data points. While research on personality traits has made significant advancements, the existence of personality types remains a subject of debate. To overcome the limitations of existing clustering methods, we applied a combination of Ward's method and k-means clustering, identifying seven distinct personality types. These types correspond to those suggested by previous studies, while also reflecting unique characteristics of Japanese culture, offering new insights into the personality structure of Japanese individuals. Additionally, the importance of the Honesty-Humility factor within the context of Japanese social norms was highlighted, illustrating the influence of cultural factors on the formation of personality types.

Background of Personality Research

Personality research is one of the longest-standing fields in psychology, with efforts to understand individual differences in behavior and personality dating back to ancient Greece (Revelle et al., 2013). Modern personality research has largely focused on the Five-Factor Model (FFM), established between the 1980s and 1990s (Goldberg, 1990; McCrae & Costa, 1997). This model seeks to describe individual personality using five major dimensions: Extraversion, Agreeableness, Conscientiousness, Neuroticism, and Openness to Experience.

Since the establishment of the FFM, personality research has expanded to integrate insights from genetics, neuroscience, and evolutionary psychology, aiming to uncover the biological basis of personality traits. Twin studies have indicated that the heritability of personality traits ranges from 40–60% (Bouchard & Loehlin, 2001), while neuroimaging studies have linked personality traits to specific brain regions and neural circuits (DeYoung et al., 2010). From an evolutionary psychology perspective, these traits are hypothesized to have played adaptive roles in human evolution (Nettle, 2006). Personality traits are associated with a wide range of psychological constructs, including well-being and life satisfaction (Steel et al., 2008), psychopathology (Kotov et al., 2010), and cognitive abilities (DeYoung et al., 2014). These findings suggest that personality plays a crucial role in psychological adaptation and functioning. Moreover, personality traits have been found to influence various aspects of life. In the realm of academic achievement, Conscientiousness has shown the strongest positive correlation, followed by Openness (Poropat, 2009). In career-related domains, traits such as Extraversion have been associated with higher compatibility in sales and leadership roles (Judge et al., 2002). In interpersonal relationships, Agreeableness has been linked to the formation and maintenance of positive social relationships (Jensen-Campbell & Graziano, 2001). Regarding health, higher levels of Neuroticism have been identified as risk factors for various physical and mental health issues (Lahey, 2009). Thus, personality research has not only contributed to our understanding of individual differences but has also advanced knowledge of their biological underpinnings, environmental interactions, and their influence across multiple life domains. Future directions include the development of more refined measurement techniques, the adoption of longitudinal study designs, and the promotion of cross-cultural comparative research.

The HEXACO Model

The HEXACO model is a relatively new personality model proposed by Ashton and Lee in the early 2000s (Ashton & Lee, 2007). This model expands upon the traditional Big Five model, including six major dimensions: Honesty-Humility, Emotionality, Extraversion, Agreeableness, Conscientiousness, and Openness to Experience. The acronym "HEXACO" is derived from the initials of these six factors. One of the most distinctive features of the HEXACO model is the addition of the Honesty-Humility factor, which includes traits related to sincerity, fairness, modesty, and low greed. By incorporating this factor, the HEXACO model allows for a clearer description of moral and ethical aspects of personality, which were not adequately captured by the Big Five (Lee & Ashton, 2004). A key difference between the HEXACO and Big Five models lies in the reconceptualization of Agreeableness and Neuroticism (Emotionality in the HEXACO model). HEXACO's Agreeableness focuses on traits such as tolerance and flexibility, characterized by the absence of anger, while Emotionality includes fearfulness, anxiety, dependence, and sentimentality, differentiating it somewhat from the Big Five's Neuroticism (Ashton et al., 2014). The HEXACO model has been shown to outperform the Big Five in explaining traits associated with the Dark Triad (Machiavellianism, psychopathy, and narcissism). For example, the Honesty-Humility factor has demonstrated a strong negative correlation with these dark traits (Lee & Ashton, 2005). Furthermore, the HEXACO model has exhibited superior predictive power in relation to socially significant behaviors and attitudes, such as counterproductive workplace behavior (Lee et al., 2005), risk-taking (de Vries et al., 2009), and substance use (Ashton & Lee, 2008). The validity of the HEXACO model has also been supported by studies across different languages and cultures. Ashton et al. (2004) analyzed lexical studies in 12 different languages, consistently identifying the six-factor structure of the HEXACO model, suggesting that it reflects a universal personality structure across cultures. In addition, the HEXACO model is supported by evolutionary psychology. In particular, the Honesty-Humility factor is thought to be associated with the evolution of cooperative behavior in humans (Ashton & Lee, 2007). Individuals high in Honesty-Humility tend to prioritize long-term reciprocal cooperation over short-term self-interest, which is seen as an adaptive trait in human societies. The HEXACO model brings new perspectives to personality research, enabling a more comprehensive understanding of human behavior and attitudes. Future research challenges include elucidating the neurobiological foundations of the model, exploring developmental changes through longitudinal studies, and expanding cross-cultural comparisons. Further investigation into the utility of the HEXACO model in applied fields such as clinical and organizational psychology is also warranted.

Current Research on Personality Types

In personality research, two main methodological approaches exist: the variable-centered approach and the person-centered approach. The variable-centered approach treats individual personality traits independently, focusing on their continuous variation. In contrast, the person-centered approach seeks to identify patterns or types of individuals characterized by specific combinations of multiple traits (Asendorpf, 2015). Research on personality types follows the latter approach, aiming to understand individuals holistically.

The history of personality type research dates back to the ancient Greek theory of the four humors, but its modern foundations were laid by Block and Block's (1980) theory of ego control and ego resiliency. Based on this theory, Robins et al. (1996) proposed three basic personality types: resilient, overcontrolled, and undercontrolled. These types have been replicated in many subsequent studies, becoming known as the ARC types (Asendorpf, Robins, Caspi). Resilient individuals exhibit low Neuroticism and high levels of other traits, reflecting adaptability to the environment. Overcontrolled individuals are characterized by high Neuroticism and low Extraversion, showing strong inhibition of emotions and impulses. Undercontrolled individuals, on the other hand, display high Extraversion and low Agreeableness and Conscientiousness, showing impulsive and less rule-bound behavior. Recent research has re-examined personality types using larger datasets and more advanced statistical methods. Daljeet et al. (2017), using latent profile analysis with the HEXACO model, identified five personality types: socially considerate, confident, goal-oriented, withdrawn, and maladaptive. Espinoza et al. (2020), in a study involving over 270,000 participants, also identified five optimal personality profiles: egocentric, flexible, disciplined, sociable, and role model types. Sava and Popa (2011), conducting a cluster analysis on a Romanian sample, identified five personality types, including resilient, overcontrolled, and undercontrolled types, as well as average and reserved types. In Japan, Kase et al. (2018), based on Big Five traits, identified four personality types, suggesting the existence of a "Not Identifiables" type unique to Japan, characterized by low Extraversion and high Agreeableness. Gerlach et al. (2018) reported four distinct personality types in a study involving over 1.5 million participants: average, role model, egocentric, and reserved types. Kerber et al. (2021), reviewing the literature and using a large German dataset, concluded that five personality types (resilient, overcontrolled, undercontrolled, reserved, and vulnerable-resilient) were optimal. These existing studies highlight several challenges. First, the choice of clustering algorithms can significantly impact results. Many studies use only a single algorithm, but comparing multiple algorithms can yield more robust results. Second, the methods for evaluating cluster solution validity remain a concern. It is important to consider both internal and external validity and use multiple indicators for comprehensive evaluation. Third, much of the existing research is based on Western cultural data, raising the possibility that cultural diversity is underrepresented. Cross-cultural comparisons are needed to clarify the universality and specificity of personality types. Finally, studies using the HEXACO model for personality type research are still relatively rare, particularly in non-Western cultures. Further research is needed to investigate how the HEXACO model's Honesty-Humility factor contributes to the formation of personality types. Addressing these challenges can help establish a stronger scientific foundation for personality type research and contribute to the advancement of individual differences psychology.

The Relationship Between Culture and Personality

The relationship between culture and personality is an important and complex theme in psychological research. The universality hypothesis of personality traits posits that the basic structure of personality is common across cultures (McCrae & Costa, 1997). This hypothesis is supported by findings that major personality models, such as the Big Five and HEXACO, are consistently replicated across different

cultural contexts (Ashton & Lee, 2007). However, while the universality of personality structures is supported, cultural differences cannot be ignored. Significant differences in the expression and evaluation of personality traits are observed between Western and Eastern cultures. For example, Western cultures tend to emphasize individualistic values, where self-assertion and individual expression are prioritized, whereas Eastern cultures, including Japan, tend to emphasize collectivist values, where relationships with others and harmony are prioritized (Markus & Kitayama, 1991). These cultural differences may influence the manifestation and interpretation of personality traits. In the context of Japanese culture, several distinctive trends have been reported. Schmitt et al. (2008), in a large-scale cross-cultural study involving 56 countries, found that compared to other nations, Japanese individuals exhibited the following traits:

- High Neuroticism: Japanese individuals tend to experience negative emotions such as anxiety and depression more frequently.
- Low Conscientiousness: Japanese individuals scored lower in aspects of planning and self-discipline compared to other nations.
- Low Extraversion: Japanese individuals tend to be more reserved and less sociable.
- Low Agreeableness: Japanese individuals scored slightly lower in cooperation and altruism.
- Low Openness: Japanese individuals tended to be more conservative and less open to new experiences.

These traits are closely related to Japanese cultural values and social norms. Notably, the high level of Neuroticism has been confirmed in studies by Kase et al. (2018), who suggested the existence of a "Not Identifiables" type unique to Japan, characterized by low Extraversion and high Agreeableness. Furthermore, Jonason et al. found that Japan exhibited higher tendencies of Machiavellianism and psychopathy, two of the Dark Triad traits, compared to other countries. While this may seem contradictory to traditional stereotypes of Japanese culture, these findings may reflect the complex social environment in Japan, where competitive pressures and the pursuit of self-interest, while maintaining outward harmony, play a role in the elevated presence of these traits. Personality research based on the HEXACO model in Japan is still limited, but the Honesty-Humility factor may offer particularly intriguing insights within the context of Japanese culture. The contrast between superficial humility and the relatively higher levels of Machiavellianism in actual behavior may provide new insights into this factor’s role. However, the accumulation of HEXACO data on Japanese individuals remains insufficient. To further explore the relationship between culture and personality, it is essential to conduct additional large-scale studies using the HEXACO model in Japan. Understanding the relationship between culture and personality is vital in today’s increasingly globalized society. To promote mutual understanding and effective communication with people from different cultural backgrounds, it is necessary to consider both the universal and culture-specific aspects of personality. Future research should aim to unravel the complex interaction between culture and personality using more sophisticated measurement methods and analytical techniques.

Purpose and Significance of the Present Study

The primary aim of this study is to identify personality types using a large dataset of Japanese individuals based on the HEXACO model and to clarify the characteristics of these types. To achieve this goal, we established the following three research questions:

Research Question 1

Do the number and characteristics of personality types derived from the Japanese HEXACO data align with those reported in previous studies? This research question seeks to examine the extent to which the personality structure of Japanese individuals aligns with the results of previous studies centered around Western cultures. Studies such as those by Daljeet et al. (2017) and Espinoza et al. (2020) have identified five personality types based on the HEXACO model. In this study, we will investigate whether the number of types extracted from the Japanese data matches the results of prior studies and whether the characteristic profiles of each type are similar. This comparison is expected to provide new insights into the universality and cultural specificity of personality types.

Research Question 2

Do the characteristics of the extracted personality types reflect the cultural characteristics of Japan? This question explores how the personality types of Japanese individuals reflect unique features of Japanese culture. According to Schmitt et al. (2008), Japanese individuals exhibit higher levels of Neuroticism and lower levels of Conscientiousness, Extraversion, Agreeableness, and Openness compared to other nations. Kase et al. (2018) also suggested the existence of a unique "Not Identifiables" type in Japan. Furthermore, Jonason et al. reported higher tendencies of Machiavellianism and psychopathy among Japanese individuals. We will analyze how the personality types extracted in this study reflect these cultural tendencies. This analysis is expected to deepen our understanding of the interaction between culture and personality.

Research Question 3

How do the six factors of the HEXACO model contribute to the formation of personality types in Japanese individuals? This research question examines the role of the six factors of the HEXACO model, particularly the Honesty-Humility factor, in shaping the personality types of Japanese individuals. As a unique factor not included in the Big Five model, the Honesty-Humility factor may play a critical role in understanding the emphasis on humility in Japanese culture, as well as the apparent contradiction between superficial humility and high levels of Machiavellianism. By analyzing the contribution of each factor, we aim to gain a more detailed understanding of the personality structure of Japanese individuals.

Methods

Data Collection

Data were collected through an online survey from July 1 to September 30, 2024. Participants completed the 60-item Japanese version of the HEXACO-PI-R (Wakabayashi, 2014). Responses were only accepted if all items were answered, resulting in valid data from 13,668 participants. The survey was conducted in compliance with the conditions stipulated by hexaco.org, which include ensuring that the survey is not indexed by search engines and is accessible only via a password-protected form. The HEXACO-PI-R consists of six domains, each subdivided into four facets:

- Honesty-Humility: Sincerity, Fairness, Greed Avoidance, Modesty
- Emotionality: Fearfulness, Anxiety, Dependence, Sentimentality
- Extraversion: Social Self-Esteem, Social Boldness, Sociability, Liveliness
- Agreeableness: Forgiveness, Gentleness, Flexibility, Patience
- Conscientiousness: Organization, Diligence, Perfectionism, Prudence
- Openness to Experience: Aesthetic Appreciation, Inquisitiveness, Creativity, Unconventionality

Participants rated each item on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree). Only anonymous response data were collected; demographic information such as gender and age was not obtained.

Descriptive Statistics

We calculated descriptive statistics, including median, mean, standard deviation, variance, skewness, and kurtosis for each domain and facet (Table 1). These statistics provide key insights into the distribution of personality traits among the sample. For example, the median and mean scores for Honesty-Humility and Emotionality were around 3.30, suggesting a relatively consistent pattern in these traits among Japanese participants. Standard deviations were .79 and .89, indicating moderate variability across participants, but without extreme deviations. In contrast, the median score for Extraversion was 2.70, with a mean of 2.80, suggesting lower levels of extraversion within the sample. The standard deviation of .91 indicated greater variability in extraversion compared to other traits. The median score for Openness to Experience was 3.70, with a mean of 3.62, which was notably higher than other domains, suggesting that Japanese participants exhibit a relatively open attitude toward new experiences and creative activities. The skewness and kurtosis values further revealed the shape of the trait distributions. For example, Extraversion and Emotionality exhibited flatter distributions with kurtosis values of − .64 and − .56, respectively. Meanwhile, Honesty-Humility showed a slightly left-skewed distribution with a skewness of − .26. The histograms for each domain provide a visual representation of the distribution of scores (Fig. 1). Honesty-Humility and Emotionality were clustered around mid-range values, suggesting that participants generally exhibited balanced attitudes in these traits. In contrast, Extraversion scores were more concentrated in the lower range, indicating that many participants scored low on extraversion. This distribution highlights a common tendency toward introversion among the Japanese sample.

Correlation and Reliability of HEXACO Traits

We examined the correlations between HEXACO traits using Spearman's rank correlation coefficients, after confirming normality using the Shapiro-Wilk test (Table 2). Several notable relationships emerged. First, there was a negative correlation between Honesty-Humility and Extraversion ($r = -.2313$), indicating that individuals with higher levels of honesty and humility tend to score lower on extraversion, possibly reflecting the influence of Japanese cultural norms that emphasize modesty and restraint. Conversely, Honesty-Humility was positively correlated with Conscientiousness ($r = .1835$), suggesting that individuals who are honest and humble are also likely to be more organized and disciplined. This relationship may reflect the close connection between integrity and responsibility in Japanese culture. In terms of reliability, Cronbach's alpha was calculated for the HEXACO scales. The overall scale ha

d an alpha of .64, indicating moderate internal consistency. Domain-specific alphas were: Honesty-Humility (.57), Emotionality (.79), Extraversion (.79), Agreeableness (.68), Conscientiousness (.76), and Openness (.72). While Emotionality and Extraversion demonstrated strong internal consistency, Honesty-Humility had a relatively low alpha of .57, suggesting that this domain may require further refinement for enhanced reliability in future research.

Derivation of the Prototypes

In this study, we followed the approach of Kerber et al. (2021), employing multiple validity criteria and clustering algorithms. Specifically, we used three different clustering techniques: a) Ward's method followed by k-means clustering (hierarchical clustering), b) latent profile analysis (LPA), and c) spectral clustering. The Ward's method and k-means clustering combination is a widely used approach in personality type research. It begins with hierarchical clustering via Ward's method, which is then used as the initial values for k-means clustering. This method reduces the risk of local optima and allows for efficient clustering. Latent profile analysis (LPA) is a probabilistic finite mixture modeling technique. It assumes that latent classes or profiles exist behind the observed variables and calculates the probability that each individual belongs to a given class. This method allows for more flexible modeling compared to traditional methods and is particularly suitable for continuous variables. Spectral clustering uses graph theory to detect non-linearly separable clusters. It constructs a similarity graph among data points and uses the eigenvectors of the graph's Laplacian matrix to perform clustering. This method is useful for identifying clusters with complex shapes that might not be captured by linear approaches. By employing these three methods, we compared clustering results across different approaches to identify robust personality types.

Determining the Number of Clusters

Determining the optimal number of clusters is a crucial step in clustering analysis. In this study, we used both internal and external validity indices to decide the optimal number of clusters. For external validity, we used Cohen's kappa and the Hubert-Arabie adjusted Rand index (ARI) to evaluate the agreement

between different clustering results. A support vector machine (SVM) classifier was also used to assign clusters, allowing for more flexible cluster assignment that accommodates non-spherical distributions. For internal validity, we employed the following indices: 1) the elbow method, 2) cost function-based criteria (AIC and BIC), 3) cluster density-based criteria (S_Dbw index), and 4) geometric assumption-based indices (C-index, Generalized Dunn index, Baker & Hubert Gamma index, G(+) index, and Silhouette index). These diverse indices provided a comprehensive evaluation of the validity of the cluster solutions.

Procedure

Following Kerber et al. (2021), the following procedure was used to identify and evaluate personality types:

1. Ten random subsamples were created from the full dataset (N = 22,820), and each subsample was split into two halves.
2. Three clustering algorithms (Ward's method + k-means, LPA, and spectral clustering) were applied to each half, yielding 3-cluster, 5-cluster, and 7-cluster solutions.
3. SVM classifiers were used to reassign the other half of the subsample to clusters.
4. Cohen's kappa and the adjusted Rand index (ARI) were calculated to evaluate the consistency of the clustering results.
5. The results were averaged across the 10 subsamples to obtain an overall evaluation for each algorithm and cluster number.
6. Internal validity indices (described in section 2.7) were calculated to assess the internal consistency of the cluster solution.
7. The results of the external and internal validity indices were combined to determine the optimal number of clusters and clustering algorithm.
8. The full dataset was clustered using the selected optimal method and number of clusters.
9. Labeling of each cluster was conducted based on the characteristics of the profiles in reference to previous studies.

This procedure enabled us to identify and evaluate HEXACO personality types in the Japanese sample. The next section provides a detailed report of the results.

Results

Internal and External Cluster Fit Indices

In evaluating the clustering solutions, both internal and external validity indices were used to assess the fit of various clustering methods. Results from the elbow method (Table 3) showed that the cost function decreased gradually as the number of clusters increased across all methods. However, the reduction became less pronounced at a certain point, which is typically used to determine the optimal

number of clusters. Specifically, for Ward's method followed by k-means clustering, there was a significant reduction in the cost function up to seven clusters (47505.87 → 43827.07), after which the reduction slowed. Similarly, both the Gaussian Mixture Model (LPA) and spectral clustering also showed a clear turning point at seven clusters. For external validity, Cohen's kappa and the Hubert-Arabie adjusted Rand index (ARI) were used to evaluate the consistency of clustering results (Table 4). Internal validity indices such as AIC, BIC, S_Dbw, C-index, Generalized Dunn index, and Silhouette index were also calculated (Table 4).

Internal Validity Indices (Table 4): The C-index showed a decrease in value as the number of clusters increased across all methods. Particularly for the 7-cluster solution, Ward's method followed by k-means showed a value of .21, Gaussian Mixture Model (GMM) showed .25, and spectral clustering showed .23, indicating that the 7-cluster solution had better clustering performance than other solutions. The Generalized Dunn Index 31 showed lower values for the 3-cluster solution (.015–.019), but slightly higher values for the 5-cluster and 7-cluster solutions (5-cluster: .018–.022, 7-cluster: .018–.022). Notably, the Ward's method followed by k-means showed the best performance for the 7-cluster solution. Results from the Baker-Hubert Gamma index and the G(+) index also indicated that the 7-cluster solution using Ward's method followed by k-means performed the best. For the Baker-Hubert Gamma index, Ward's method followed by k-means yielded a value of .55, GMM showed .47, and spectral clustering showed .52. For the G(+) index, Ward's method followed by k-means achieved a high score of .70. The Silhouette index showed relatively lower results overall, with Ward's method followed by k-means showing the highest value of .144 for the 3-cluster solution. Other methods showed values ranging from .080 to .111, with Ward's method + k-means exhibiting relatively better clustering performance. The S_Dbw index values ranged from 1.61 to 1.75 across all methods, with a slight increase as the number of clusters increased. The GMM 3-cluster solution showed the lowest value of 1.61.

External Validity Indices (Table 4)

For Cohen's kappa, the 7-cluster solution using Ward's method followed by k-means showed the highest value (.9388), with GMM also yielding a high consistency score (.9050). In contrast, spectral clustering produced a low score of − .0011 for the 7-cluster solution, indicating that Ward's method followed by k-means and GMM were the most reliable clustering methods for this solution. The Adjusted Rand Index (ARI) also showed strong results for the 7-cluster solution using Ward's method followed by k-means (.8829) and GMM (.8264). In contrast, spectral clustering yielded a lower ARI of .3817, further supporting the superior performance of Ward's method followed by k-means and GMM.

In conclusion, the results of both external and internal validity indices indicated that the 7-cluster solution using Ward's method followed by k-means provided the best overall performance. Cohen's kappa and ARI scores were particularly high, confirming the consistency and stability of this clustering method. GMM also performed well in the 7-cluster solution, suggesting it as another viable clustering method.

Clustering and Labeling

Using the 7-cluster solution from the k-means method, the average scores of each cluster on the HEXACO traits were calculated (Fig. 2). Additionally, the percentage of participants in each cluster was determined, and the clusters were labeled based on their profiles.

- Cluster 1: Honesty-Humility (3.3), Emotionality (2.7), Extraversion (2.44), Agreeableness (2.87), Conscientiousness (2.41), Openness (2.48). This cluster is characterized by introversion, agreeableness, and slightly lower self-management skills. Comprising 13.81% of the sample, it was labeled as the “Reserved Type.” Individuals in this group are modest and prioritize social harmony but may face challenges in self-management.
- Cluster 2: Honesty-Humility (3.0), Emotionality (3.93), Extraversion (2.16), Agreeableness (2.98), Conscientiousness (2.41), Openness (3.62). This group exhibits high emotional sensitivity and low extraversion but is highly open-minded. Representing 16.96% of the sample, it was labeled as the “Emotionally Sensitive Type.” Individuals here express emotions openly and possess a strong sense of creativity and openness to new experiences.
- Cluster 3: Honesty-Humility (3.82), Emotionality (3.9), Extraversion (2.06), Agreeableness (3.15), Conscientiousness (3.72), Openness (3.84). This cluster consists of highly conscientious individuals who are also introverted and exhibit strong intellectual curiosity. With 17.3% of the sample, it was labeled as the “Conscientious Introvert Type.” These individuals demonstrate high levels of honesty and responsibility, with a preference for solitary and reflective activities.
- Cluster 4: Honesty-Humility (3.36), Emotionality (3.68), Extraversion (3.67), Agreeableness (3.55), Conscientiousness (3.06), Openness (3.85). Individuals in this cluster are sociable, agreeable, and excel in social interactions. Representing 15.47% of the sample, this group was labeled the “Extraverted Agreeable Type.” These individuals are outgoing, cooperative, and socially engaged.
- Cluster 5: Honesty-Humility (2.79), Emotionality (2.91), Extraversion (3.09), Agreeableness (2.2), Conscientiousness (3.78), Openness (3.81). This group is characterized by high self-management but lower agreeableness. Representing 11.85% of the sample, this cluster was labeled as the “Independent Type.” These individuals exhibit strong self-discipline and independence but may struggle with maintaining harmonious relationships with others.
- Cluster 6: Honesty-Humility (2.36), Emotionality (2.6), Extraversion (3.94), Agreeableness (2.86), Conscientiousness (2.32), Openness (3.66). This group is sociable and emotionally stable but exhibits lower self-management skills. With 11.64% of the sample, this cluster was labeled as the “Extraverted Stable Type.” These individuals are confident and emotionally composed, with high social energy.
- Cluster 7: Honesty-Humility (3.79), Emotionality (2.3), Extraversion (2.68), Agreeableness (3.27), Conscientiousness (2.91), Openness (4.06). This group is introverted but highly conscientious and open to new experiences, often demonstrating leadership skills. Representing 12.97% of the sample, this group was labeled as the “Introverted Leader Type.” These individuals are honest, flexible thinkers, and possess leadership potential despite their introverted nature.

Each of these clusters reflects distinct personality profiles based on the HEXACO model and highlights the diversity of personality types within the Japanese sample. These findings offer a nuanced understanding of individual personality tendencies within the context of Japanese culture.

Discussion

Summary of Results

This study clustered over 13,000 personality data points from Japanese individuals based on the HEXACO model, identifying seven distinct personality types. The combination of Ward's method and k-means clustering showed the best fit in terms of both internal and external validity, with particularly high scores for Cohen's kappa and the adjusted Rand index. The seven identified clusters are as follows:

1. Reserved Type: Introverted, agreeable, but with lower self-management skills.
2. Emotionally Sensitive Type: Emotionally sensitive, low extraversion, but highly open-minded.
3. Conscientious Introvert Type: Honest, conscientious, introverted individuals with high responsibility.
4. Extraverted Agreeable Type: Outgoing and highly cooperative in social settings.
5. Independent Type: Self-disciplined and independent, with lower agreeableness.
6. Extraverted Stable Type: Sociable and emotionally stable but lower in self-management.
7. Introverted Leader Type: Honest, introverted, and open to new experiences, with leadership potential.

These clusters reflect various combinations of HEXACO traits, highlighting the diversity of personality types within Japanese individuals. Some of the clusters align with previously reported personality types, while others reveal characteristics unique to Japanese culture.

Interpretation of Findings

The results of this study provide novel insights into personality types in Japan, aligning with some aspects of previous research while also identifying personality types that are specific to Japanese culture. For example, the Reserved Type and Conscientious Introvert Type correspond to overcontrolled and resilient types reported in previous studies. However, these types also reflect a strong emphasis on introversion and social harmony, which are more pronounced in Japanese culture. The Emotionally Sensitive Type and Introverted Leader Type also stand out, reflecting a greater sensitivity to emotional nuances and a more reserved form of leadership that is valued in Japan. These findings support the research questions proposed in section 1.5, particularly the first question regarding whether Japanese personality types align with those identified in previous studies. While the results confirm the existence of universal personality structures, the seven clusters identified in this study highlight unique cultural traits that differentiate Japanese individuals from those in Western studies.

Cultural Influences on Personality

The findings of this study reaffirm the importance of cultural factors in shaping personality types, as highlighted in previous personality research. While some personality types, such as the Resilient and Overcontrolled types, were replicated in the Japanese sample, other clusters demonstrated unique cultural influences on personality formation. For instance, the Reserved Type and Conscientious Introvert Type reflect Japanese cultural norms that value introversion and modesty, traits that are culturally reinforced through societal expectations of politeness and social harmony. These traits stand in contrast to more extroverted personality profiles that are more prevalent in Western societies. On the other hand, clusters such as the Extraverted Stable Type and the Independent Type, characterized by lower levels of Agreeableness or higher levels of Extraversion, reflect a shift towards individualism in modern Japanese society, diverging from traditional collectivist norms. Additionally, the high prevalence of Emotional Sensitivity and Introverted Leadership in certain clusters highlights the nuanced role that emotional expressiveness and leadership play in Japanese culture. These traits suggest that, while emotional regulation and humility are often emphasized, there is room for creative and emotionally sensitive leadership within the Japanese social context. In summary, this study illustrates how cultural values, such as humility, social harmony, and emotional sensitivity, are embedded in Japanese personality structures, influencing the development and manifestation of personality types.

Limitations of the Study

While this study provides important insights into the personality structure of Japanese individuals using the HEXACO model, several limitations must be acknowledged. These limitations should be considered when interpreting the results and in the design of future research. First, the lack of demographic data in this study, such as gender, age, and geographic region, restricts the ability to examine how these factors may influence the formation and characteristics of personality types. Including such demographic information could provide a more granular understanding of personality types and their development across different population subgroups. Future studies could address this limitation by analyzing how personality types vary across demographic categories within the Japanese population. Second, this study did not examine the relationships between personality types and other important psychological or behavioral variables, such as physical and mental health, well-being, self-esteem, or risk-taking behavior. These variables are known to be closely linked to personality traits and could offer a deeper understanding of how personality types affect various life outcomes. Future research could incorporate these factors to develop a more comprehensive picture of personality types and their real-world implications. Third, the internal consistency of the Honesty-Humility domain, as indicated by its relatively low Cronbach's alpha (.57), suggests that this domain may require further refinement. This raises the question of whether the measurement of this domain accurately captures the nuances of Honesty-Humility in Japanese culture. Future studies could improve the scale's reliability by refining the items or adapting them to better fit the cultural context. Additionally, the cross-cultural comparison of HEXACO personality types remains a future research challenge. While this study focused solely on Japanese individuals, comparing these findings with data from other non-Western and Western cultures would provide more insight into the universality and cultural specificity of personality types. Such comparative studies could help clarify how cultural norms and values influence personality structures across different

societies. Finally, the study's use of an internet-based survey may have introduced sampling biases, potentially over-representing individuals who are more comfortable using digital platforms. This could affect the generalizability of the findings to the broader Japanese population. Future studies should consider more diverse data collection methods to obtain a representative sample. Despite these limitations, this study contributes significantly to understanding Japanese personality structures based on the HEXACO model and highlights the cultural factors that influence personality formation.

Implications for Future Research

The findings of this study offer several directions for future research. First, future studies should aim to address the limitations mentioned above, particularly by incorporating demographic variables and behavioral indicators such as health and well-being. This would help clarify the role of personality types in different life domains and provide more practical applications for the findings. Moreover, cross-cultural studies using the HEXACO model would enable researchers to explore how cultural contexts shape personality types. This could lead to a more comprehensive understanding of the interplay between universal personality traits and cultural values, thus enriching the field of personality research. Lastly, the refinement of the Honesty-Humility domain within the Japanese context is an important future research direction. As this study indicated some reliability issues with this domain, further investigations into how it can be more accurately measured and understood in non-Western contexts will be essential for advancing the HEXACO model's applicability.

Conclusion

This study identified seven distinct personality types in a sample of Japanese individuals based on the HEXACO model. These clusters offer new insights into the structure of Japanese personalities, while also reflecting both universal and culturally specific aspects of personality. Particularly notable was the influence of the Honesty-Humility factor, which plays a significant role in shaping personality types within the context of Japanese social norms, as well as the relatively lower levels of Extraversion and Agreeableness observed in this sample. These findings highlight the importance of cultural factors in personality formation and provide a valuable foundation for future research into personality types in non-Western contexts. The identified personality types can also be applied in fields such as clinical psychology, education, and organizational management, helping to develop tailored support systems and management strategies that align with the unique personality profiles observed in Japanese individuals. Moving forward, further exploration of cross-cultural differences and similarities in personality types, as well as an examination of the relationships between personality traits and psychological outcomes such as well-being and risk-taking behavior, will enrich our understanding of personality in a globalized world. This study serves as a step toward such comprehensive research, with implications for both theory and practice.

Declarations

Ethics Approval and Consent to Participate

This study was conducted in accordance with the ethical guidelines for medical and health research involving human subjects set by Japan's Ministry of Health, Labour and Welfare. Informed consent was obtained from all participants. To protect participant privacy, responses were collected anonymously, and participants were assured that any requests for correction or deletion of their information would be handled promptly.

Consent for publication

Informed consent for the publication of the research results was obtained from all participants involved in this study.

Competing interests

The author declares that there are no financial or personal relationships with other people or organizations that could inappropriately influence (bias) this work. There are no conflicts of interest to declare.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Ethics Statement This study did not receive approval from a specific ethics committee, as most ethics approval institutions in Japan are affiliated with universities, and external organizations, such as private companies, typically do not conduct ethics reviews for psychology research. After consulting with the Research and Development Division of the Health Policy Bureau at the Ministry of Health, Labour, and Welfare, I was informed that no institutions are known to conduct ethics reviews for external organizations in this field. Additionally, after explaining the content of my research, the Ministry confirmed that this study does not pose any issues under the "Ethical Guidelines for Medical and Health Research Involving Human Subjects."

This research was conducted in accordance with these ethical guidelines. Informed consent was obtained from all participants, with clear information provided on the study's purpose, data collection methods, privacy protection measures, voluntary nature of participation, and the option to withdraw at any time. To protect participant privacy, data were collected anonymously, securely stored, and participants were informed of their right to request correction or deletion of their data.

References

1. Akaike H (1998) Information Theory and an Extension of the Maximum Likelihood Principle. In: Parzen E, Tanabe K, Kitagawa G (eds) Selected Papers of Hirotugu Akaike. Springer Series in Statistics. Springer, New York, NY. https://doi.org/10.1007/978-1-4612-1694-0_15

2. Ashton MC, Lee K (2007) Empirical, theoretical, and practical advantages of the HEXACO model of personality structure. Personality Social Psychol Rev 11(2):150–166. https://doi.org/10.1177/1088868306294907
3. Ashton MC, Lee K, De Vries RE (2014) The HEXACO Honesty-Humility, Agreeableness, and Emotionality factors: A review of research and theory. Personality Social Psychol Rev 18(2):139–152. https://psycnet.apa.org/doi/10.1177/1088868314523838
4. Asendorpf JB (2015) Person-centered approaches to personality. In M. Mikulincer, P. R. Shaver, M. L. Cooper, & R. J. Larsen (Eds.), APA handbook of personality and social psychology, Vol. 4: *Personality processes and individual differences* (pp. 403–424). American Psychological Association. https://psycnet.apa.org/doi/10.1037/14343-018
5. Baker FB, Hubert LJ (1975) Measuring the power of hierarchical cluster analysis. J Am Stat Assoc 70(349):31–38. https://doi.org/10.2307/2285371
6. Block J, Block JH (1980) The role of ego-control and ego-resiliency in the organization of behavior. In: Collins WA (ed) Development of cognition, affect, and social relations: The Minnesota symposia on child psychology, vol 13. Erlbaum, pp 39–101
7. Bouchard TJ, Loehlin JC (2001) Genes, evolution, and personality. Behav Genet 31(3):243–273. https://doi.org/10.1023/a:1012294324713
8. Daljeet KN, Bremner NL, Giammarco EA, Meyer JP, Paunonen SV (2017) Taking a person-centered approach to personality: A latent-profile analysis of the HEXACO model of personality. J Res Pers 70:241–251. https://psycnet.apa.org/doi/10.1016/j.jrp.2017.08.003
9. Dalrymple-Alford E (1970) Measurement of clustering in free recall. Psychol Bull 74(1):32. https://awspntest.apa.org/doi/10.1037/h0029393
10. de Vries RE, de Vries A, Feij JA (2009) Sensation seeking, risk-taking, and the HEXACO model of personality. Pers Indiv Differ 47(6):536–540. https://doi.org/10.1016/j.paid.2009.05.029
11. DeYoung CG, Hirsh JB, Shane MS, Papademetris X, Rajeevan N, Gray JR (2010) Testing predictions from personality neuroscience: Brain structure and the Big Five. Psychol Sci 21(6):820–828. https://doi.org/10.1177/0956797610370159
12. DeYoung CG, Quilty LC, Peterson JB, Gray JR (2014) Openness to experience, intellect, and cognitive ability. J Pers Assess 96(1):46–52. https://doi.org/10.1080/00223891.2013.806327
13. Espinoza JA, Daljeet KN, Meyer JP (2020) Establishing the structure and replicability of personality profiles using the HEXACO-PI-R. Nat Hum Behav 4(7):713–724. https://psycnet.apa.org/doi/10.1038/s41562-020-0853-6
14. Gerlach M, Farb B, Revelle W, Amaral LAN (2018) A robust data-driven approach identifies four personality types across four large data sets. Nat Hum Behav 2(10):735–742. https://doi.org/10.1038/s41562-018-0419-z
15. Goldberg LR (1990) An alternative description of personality: The Big-Five factor structure. J Personal Soc Psychol 59(6):1216–1229. https://psycnet.apa.org/doi/10.1037/0022-3514.59.6.1216

16. Hubert L, Arabie P (1985) Comparing partitions. J Classif 2(1):193–218. https://doi.org/10.1007/BF01908075
17. Bezdek JC, Pal NR (1998) Some new indexes of cluster validity, in *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, vol. 28, no. 3, pp. 301–315, June https://doi.org/10.1109/3477.678624 PMID: 18255949
18. Jensen-Campbell LA, Graziano WG (2001) Agreeableness as a moderator of interpersonal conflict. J Pers 69(2):323–362. https://psycnet.apa.org/doi/10.1111/1467-6494.00148
19. Jonason PK, Foster JD, Oshio A, Sitnikova M, Birkas B, Gouveia VV (2017) Self-construals and the Dark Triad traits in six countries. Pers Indiv Differ 113:120–124. https://doi.org/10.1016/j.paid.2017.02.053
20. Judge TA, Bono JE, Ilies R, Gerhardt MW (2002) Personality and leadership: A qualitative and quantitative review. J Appl Psychol 87(4):765–780. https://psycnet.apa.org/doi/10.1037/0021-9010.87.4.765
21. Kase T, Ueno Y, Oishi K (2018) Characteristics of Resilients, Overcontrollers, Undercontrollers, and other types in personality prototypes: An examination based on high and low life skills. Japanese J Personality 27(1):79–81. https://doi.org/10.1080/00223890701629813
22. Kerber A, Roth M, Herzberg PY (2021) Personality types revisited–a literature-informed and data-driven approach to an integration of prototypical and dimensional constructs of personality description. PLoS ONE 16(1):e0244849. https://doi.org/10.1371/journal.pone.0244849
23. Kotov R, Gamez W, Schmidt F, Watson D (2010) Linking big personality traits to anxiety, depressive, and substance use disorders: A meta-analysis. Psychol Bull 136(5):768–821. https://doi.org/10.1037/a0020327
24. Lahey BB (2009) Public health significance of neuroticism. Am Psychol 64(4):241–256. https://psycnet.apa.org/doi/10.1037/a0015309
25. Lee K, Ashton MC (2004) Psychometric properties of the HEXACO personality inventory. Multivar Behav Res 39(2):329–358. https://psycnet.apa.org/doi/10.1207/s15327906mbr3902_8
26. Lee K, Ashton MC (2005) Psychopathy, Machiavellianism, and narcissism in the Five-Factor Model and the HEXACO model of personality structure. Pers Indiv Differ 38(7):1571–1582. https://psycnet.apa.org/doi/10.1016/j.paid.2004.09.016
27. Markus HR, Kitayama S (1991) Culture and the self: Implications for cognition, emotion, and motivation. Psychol Rev 98(2):224–253. https://psycnet.apa.org/doi/10.1037/0033-295X.98.2.224
28. McCrae RR, Costa PT Jr. (1997) Personality trait structure as a human universal. Am Psychol 52(5):509–516. https://psycnet.apa.org/doi/10.1037/0003-066X.52.5.509
29. Halkidi M, Vazirgiannis M (2001) Clustering validity assessment: finding the optimal partitioning of a data set, Proceedings 2001 IEEE International Conference on Data Mining, San Jose, CA, USA, pp. 187–194, 10.1109/ICDM.2001.989517
30. Milligan GW, Mahajan V (1980) A note on procedures for testing the quality of a clustering of a set of objects. Decis Sci 11(4):669–677. https://doi.org/10.1111/j.1540-5915.1980.tb01168.x

31. Nettle D (2006) The evolution of personality variation in humans and other animals. Am Psychol 61(6):622–631. https://psycnet.apa.org/doi/10.1037/0003-066X.61.6.622
32. Poropat AE (2009) A meta-analysis of the five-factor model of personality and academic performance. Psychol Bull 135(2):322–338. https://psycnet.apa.org/doi/10.1037/a0014996
33. Revelle W, Wilt J, Condon DM (2011) Individual Differences and Differential Psychology. In The Wiley-Blackwell Handbook of Individual Differences (eds T. Chamorro-Premuzic, S. von Stumm and A. Furnham). https://doi.org/10.1002/9781444343120.ch1
34. Robins RW, John OP, Caspi A, Moffitt TE, Stouthamer-Loeber M (1996) Resilient, overcontrolled, and undercontrolled boys: Three replicable personality types. J Personal Soc Psychol 70(1):157–171. https://psycnet.apa.org/doi/10.1037/0022-3514.70.1.157
35. Rohlf FJ (1974) Methods of comparing classifications. Annu Rev Ecol Syst 5101–113. https://doi.org/10.1146/annurev.es.05.110174.000533
36. Rousseeuw PJ (1987) Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. J Comput Appl Math 20:53–65. https://doi.org/10.1016/0377-0427(87)90125-7
37. Sava FA, Popa RI (2011) Personality types based on the big five model. A cluster analysis over the Romanian population. Cognitie, Creier, Comportament/Cognition. Brain Behav 15(3):359–384
38. Schmitt DP, Allik J, McCrae RR, Benet-Martínez V (2007) The geographic distribution of Big Five personality traits: Patterns and profiles of human self-description across 56 nations. J Cross-Cult Psychol 38(2):173–212. https://psycnet.apa.org/doi/10.1177/0022022106297299
39. Steel P, Schmidt J, Shultz J (2008) Refining the relationship between personality and subjective well-being. Psychol Bull 134(1):138–161. https://psycnet.apa.org/doi/10.1037/0033-2909.134.1.138
40. Wakabayashi A (2014) A sixth personality domain that is independent of the Big Five domains: The psychometric properties of the HEXACO Personality Inventory in a Japanese sample. Jpn Psychol Res 56:211–223. https://doi.org/10.1111/jpr.12045

Figures

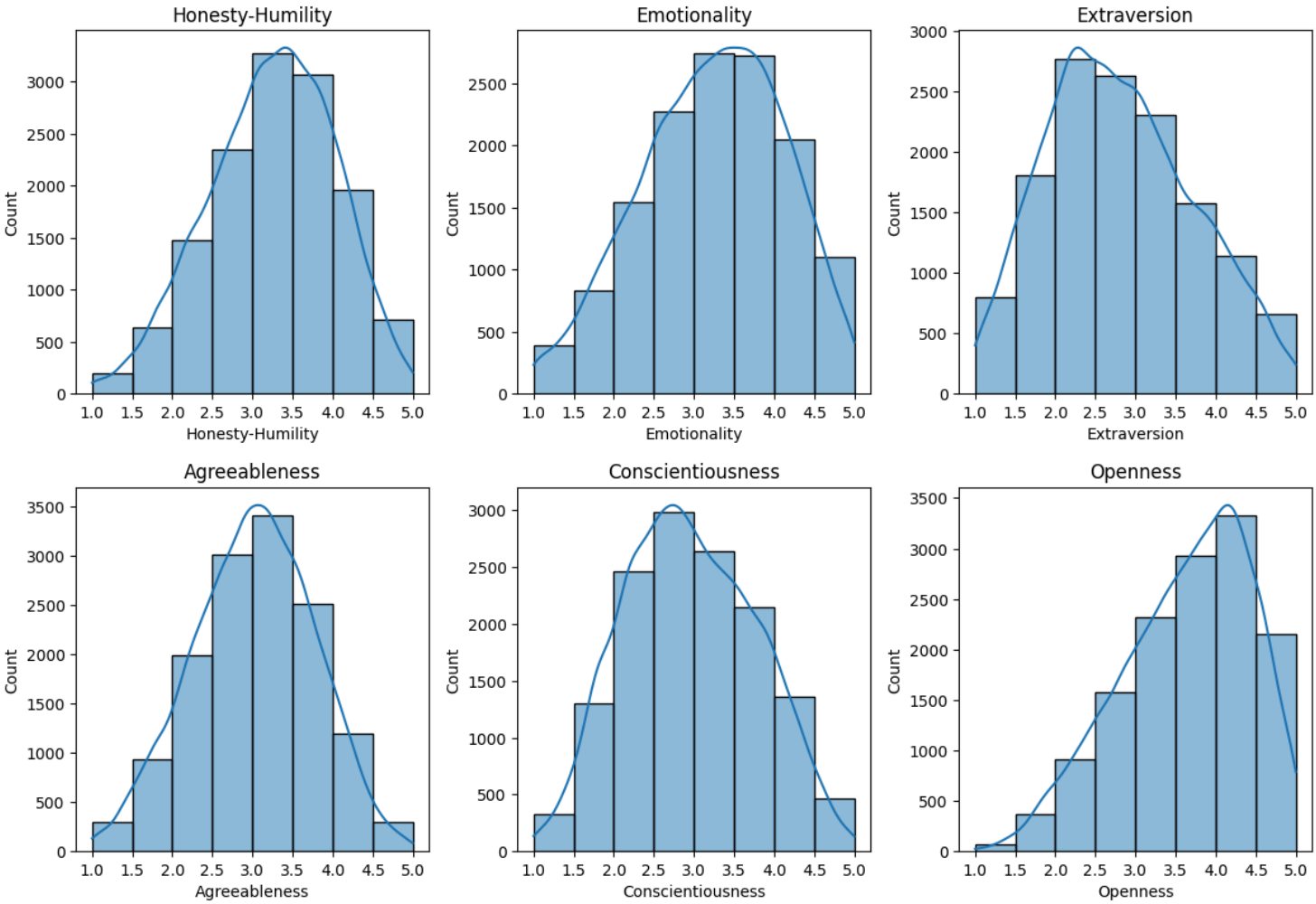


Figure 1

Legend not included with this version.

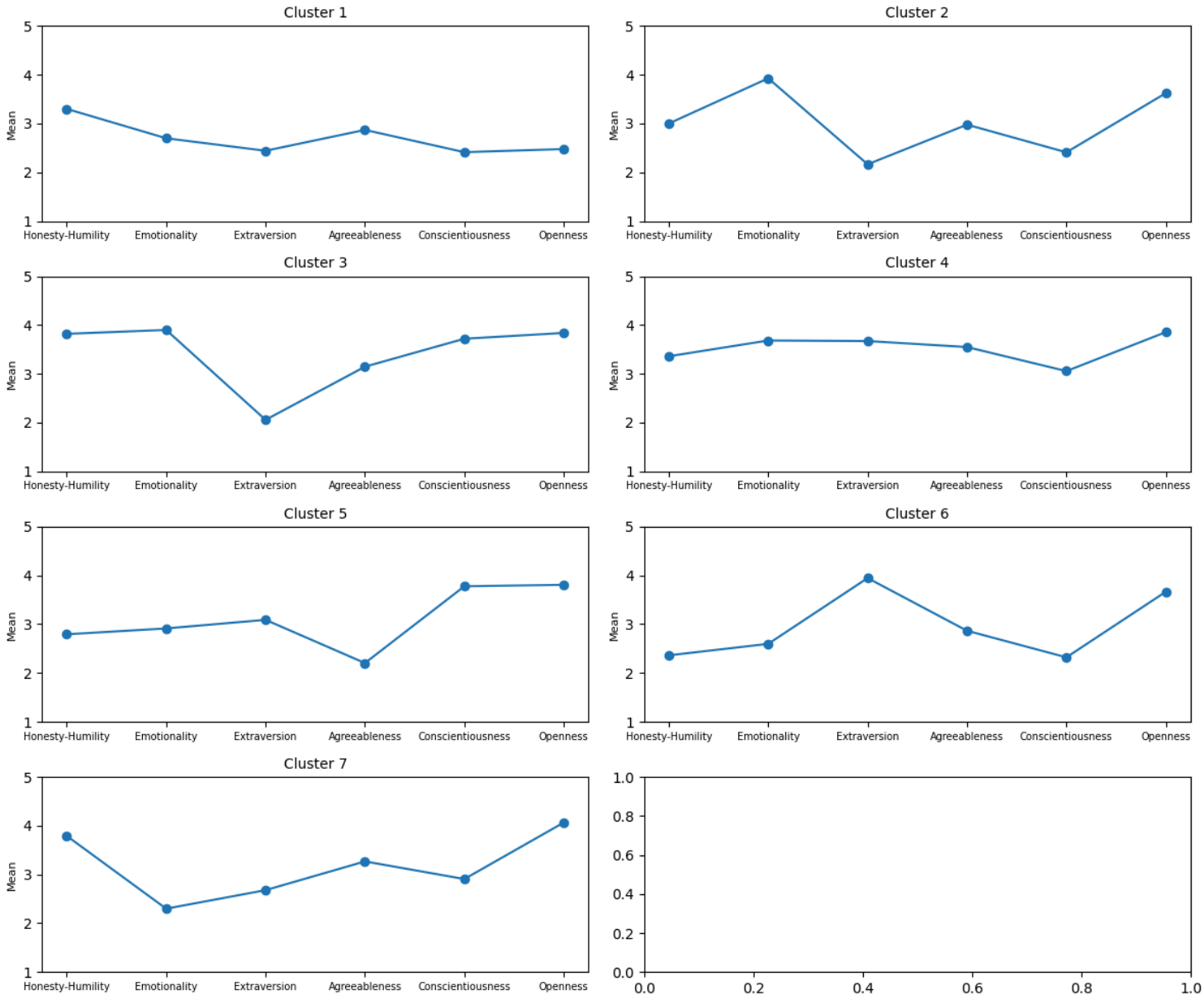


Figure 2

Legend not included with this version.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- hexacodomain.csv
- hexacofacet.csv